

Grade 6 Accelerated
Unit 6
Lesson 2 Homework

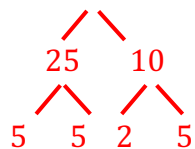
Name_____Answer Key_____

Date_____

Period_____

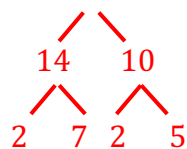
Find the prime factorization of each value.

1. 250



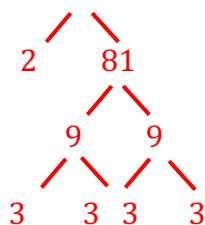
$$2 \cdot 5 \cdot 5 \cdot 5 \text{ or } 2 \cdot 5^3$$

2. 140



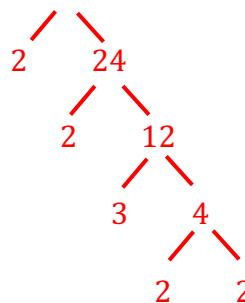
$$2 \cdot 2 \cdot 5 \cdot 7 \text{ or } 2^2 \cdot 5 \cdot 7$$

3. 162



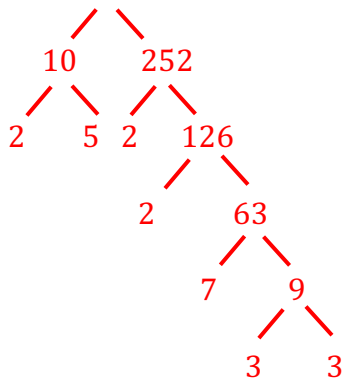
$$2 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \text{ or } 2 \cdot 3^4$$

4. 48



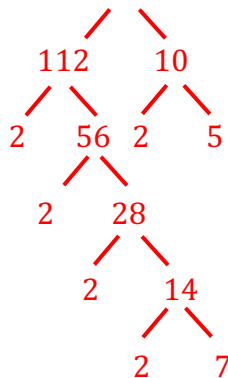
$$2 \cdot 2 \cdot 2 \cdot 2 \cdot 3 \text{ or } 2^4 \cdot 3$$

5. 2520



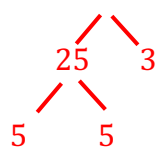
$$2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 5 \cdot 7 \text{ or } 2^3 \cdot 3^2 \cdot 5 \cdot 7$$

6. 1120



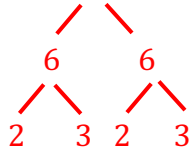
$$2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 5 \cdot 7 \text{ or } 2^5 \cdot 5 \cdot 7$$

7. 75



$$3 \cdot 5 \cdot 5 \text{ or } 3 \cdot 5^2$$

8. 36



$$2 \cdot 2 \cdot 3 \cdot 3 \text{ or } 2^2 \cdot 3^2$$

9. Find a number that meets the following conditions.
- The value of the number is less than 250.
 - One of its prime factors has an exponent of 4.
 - When expressed in exponential form, its prime factorization has 3 bases.

$$2^4 \cdot 3 \cdot 5 = 240$$

10. Find a number that meets the following conditions.
- The value of the number is greater than 900 but less than 1000.
 - One of its prime factors has an exponent of 2.
 - When expressed in exponential form, its prime factorization has 4 bases.

$$2 \cdot 3^2 \cdot 5 \cdot 11 = 990$$

or

$$2^2 \cdot 3 \cdot 7 \cdot 11 = 924$$